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Video 2 · Algebra I Worked Solutions

Cambridge IGCSE Mathematics 0580, Paper 2 Non-calculator (Extended). Full worked solutions to Questions 7–11 of the NurNabi Academy Full Revision Paper, solved the simplest way, with the method and accuracy marks shown at every step, the way a Cambridge examiner awards them.

5	17	Q7–Q11	17:50
questions	marks	E2.2 – E2.11	video runtime

M1 method mark A1 accuracy mark (needs the M mark) B1 independent mark

Question 7

Expanding three brackets

Expand and simplify $(x + 3)(x - 2)(2x + 1)$. [3]

Step 1: expand the first two brackets only:

$(x + 3)(x - 2) = x^2 - 2x + 3x - 6 = x^2 + x - 6$ **M1**

Step 2: multiply every term by the third bracket:

$(x^2 + x - 6)(2x + 1)$ **M1**

$= 2x^3 + x^2 + 2x^2 + x - 12x - 6$

Step 3: collect like terms.

$2x^3 + 3x^2 - 11x - 6$ **A1**

Second M1 is for at least 3 of the 4 final terms correct (or 5 of the 6 terms before simplifying).

Common mistakes: forgetting to multiply the -6 by *both* $2x$ and 1, or sign errors with $-2x + 3x$.

Check with $x = 1$: $(4)(-1)(3) = -12$, and $2 + 3 - 11 - 6 = -12$ ✓

Question 8

Algebraic fractions

Write $\frac{3}{x-1} - \frac{2}{x+2}$ as a single fraction in its simplest form. [3]

Common denominator: multiply the two bottoms, $(x - 1)(x + 2)$ **M1**

$\frac{3(x+2) - 2(x-1)}{(x-1)(x+2)}$ **M1**

Top: $3x + 6 - 2x + 2 = x + 8$ $(-2 \times -1 = +2)$

$$\frac{x+8}{(x-1)(x+2)}$$ **A1**

Biggest trap: writing $3x + 6 - 2x - 2$. Put **brackets** round every numerator, then the minus multiplies both terms: $-2(x - 1) = -2x + 2$.

You may leave the bottom factorised. Expanding it to $x^2 + x - 2$ is also correct.

Check with $x = 2$: $3 - \frac{2}{4} = 2.5$, and $\frac{10}{1 \times 4} = 2.5$ ✓

Question 9

Completing the square

(a) Write $x^2 - 6x + 4$ in the form $(x - a)^2 - b$. [2]

Halve the x-coefficient: $-6 \div 2 = -3$, so start with $(x - 3)^2$

$(x - 3)^2 = x^2 - 6x + 9$, which is 9 too many for the constant, so subtract it:

$x^2 - 6x + 4 = (x - 3)^2 - 9 + 4$

$(x - 3)^2 - 5$ B1 B1

B1 for $a = 3$, B1 for $b = 5$.

(b) Hence solve $x^2 - 6x + 4 = 0$. Answers in surd form. [1]

$(x - 3)^2 - 5 = 0$

$(x - 3)^2 = 5$

$x - 3 = \pm\sqrt{5}$ (don't forget the ±)

$x = 3 + \sqrt{5}$ or $x = 3 - \sqrt{5}$ FT

"Hence" means use part (a). FT: follow-through from your (a). Missing the ± loses the answer with the minus sign.

Question 9 (continued)

Completing the square

(c) Write down the coordinates of the turning point of $y = x^2 - 6x + 4$. [1]

From $(x - a)^2 - b$, the turning point is $(a, -b)$.

$(x - 3)^2 - 5 \rightarrow x = 3, y = -5$

(3, −5) FT

Why? $(x - 3)^2$ is never negative, so the smallest y can be is when $(x - 3)^2 = 0$, i.e. $x = 3$, giving $y = -5$. That minimum is the turning point.

Sign trap: the x-coordinate is +3 even though the bracket says −3.

Question 10

Changing the subject

Make x the subject of $y = \frac{x + 2}{x - 3}$. [3]

1. Clear the fraction: multiply both sides by $(x - 3)$

$y(x - 3) = x + 2 \rightarrow xy - 3y = x + 2$ M1

2. Collect the x terms on one side:

$xy - x = 3y + 2$

3. Factorise out x: $x(y - 1) = 3y + 2$ M1

4. Divide.

$$x = \frac{3y + 2}{y - 1}$$ A1

Rule: when the subject appears **twice**, you must collect the terms and **factorise**. There is no other way.

Common mistake: expanding $y(x - 3)$ as $xy - 3$. The y multiplies the 3 as well.

Also correct: $x = \frac{-3y - 2}{1 - y}$ (top and bottom × −1).

Question 11

Simultaneous equations — one linear, one non-linear

Solve $y = x + 1$ and $y = x^2 - 2x - 3$. [4]

Both equal y, so set them equal:

$x^2 - 2x - 3 = x + 1$ **M1**

Make it = 0: $x^2 - 3x - 4 = 0$

Factorise (two numbers × to −4, + to −3: −4 and +1): **M1**

$(x - 4)(x + 1) = 0$

$x = 4$ or $x = -1$

$x = 4, y = 5$ and $x = -1, y = 0$ **A1** **A1**

Find each y using the **straight line** (easier):

```
x =  4:  y =  4 + 1 = 5
x = -1:  y = -1 + 1 = 0
```

A1 for both x values, A1 for both y values, each paired correctly.

Common mistakes: stopping after finding x (lose the last A1), or pairing the wrong y with each x. Write them as pairs.

Check in the curve: $4^2 - 8 - 3 = 5$ ✓ and $1 + 2 - 3 = 0$ ✓

Answers at a glance

Video 2 · Algebra I · 17 marks

Q	Topic	Answers	Marks
7	Expanding three brackets	$2x^3 + 3x^2 - 11x - 6$	3
8	Algebraic fractions	$\frac{x + 8}{(x - 1)(x + 2)}$	3
9	Completing the square	(a) $(x - 3)^2 - 5$ (b) $3 \pm \sqrt{5}$ (c) $(3, -5)$	4
10	Changing the subject	$x = \frac{3y + 2}{y - 1}$	3
11	Simultaneous equations	(4, 5) and (−1, 0)	4
	Total		**17**

Mark allocations follow the style of the Cambridge 0580 mark scheme. The questions are original NurNabi Academy practice questions, not a prediction of the real paper. Next: **Video 3 · Algebra II (Q12–Q15)**.